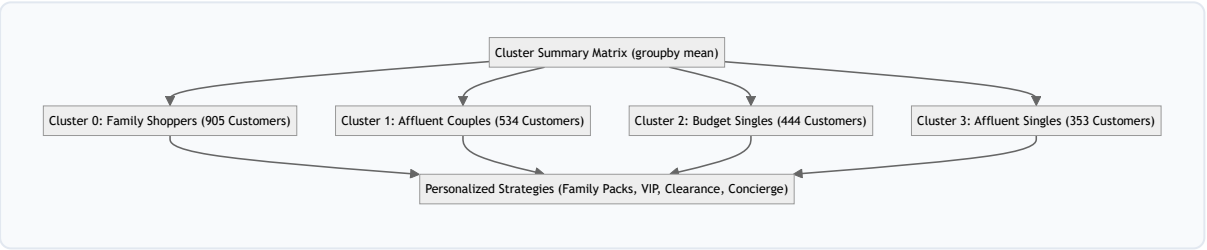
Submodule 33.12: Cluster Summary & Customer Persona Profiling (Actionable Marketing Strategy) (Hinglish)

Topic Overview & Business Translation

Unsupervised machine learning ka final goal sirf mathematical numbers nikalna nahi, balki un clusters ko real-world **Customer Personas** aur commercial business strategy mein translate karna hota hai.

Feature matrix ko cluster label se `groupby` karke mean calculate karne par **Cluster Summary Table** generate hota hai.



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1. Complete Cluster Summary Table

Reference: smartcart.ipynb (Cell 65).

```
# Cluster summary means display karna
cluster_summary = X.groupby("cluster").mean()
print(cluster_summary)
```

Exact Feature Means Har Cluster ke liye:

| Feature | Cluster 0 (Red) | Cluster 1 (Blue) | Cluster 2 (Yellow) | Cluster 3 (Green) | | :--- | :---: | :---: | :---: | :---: | | **Customer Count** | **905** | **534** | **444** | **353** | | **Income ($)** | **\$39,681** | **\$72,808** | **\$36,960** | **$70,723** | | **Total Spending ($)** | **\$222** | **\$1,237** | **\$166** | **$1,190** | | **Total Children** | **1.24** | **0.51** | **1.27** | **0.46** | | **Living_With_Partner** | **1.00 (100%)** | **1.00 (100%)** | 0.007 (~0%) | **0.00 (0%)** | | **Living_With_Alone** | **0.00 (0%)** | **0.00 (0%)** | **0.993 (~100%)** | **1.00 (100%)** | | **Customer Age** | 55.7 | 59.5 | 55.7 | 58.9 | | **Customer Tenure (Days)** | 342.9 | 369.7 | 338.8 | 376.3 | | **Recency (Days)** | 48.9 | 49.2 | 48.3 | 50.5 | | **NumDealsPurchases** | **2.59** | 1.96 | **2.59** | 1.86 | | **NumWebPurchases** | 3.15 | 5.69 | 2.71 | 5.79 | | **NumCatalogPurchases** |

0.97 | **5.50** | 0.84 | **5.01** | | **NumStorePurchases** | 4.14 | **8.66** | 3.62 | **8.43** | | **NumWebVisitsMonth** | **6.31** | 3.58 | **6.66** | 3.73 | | **Campaign Response Rate** | 7.6% | 16.7% | 14.2% | **32.0%** | | **Complain Rate** | 1.1% | 0.6% | 1.1% | 0.6% |

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2. Four Customer Personas & Business Playbooks

Persona 0: "Family Shoppers / Thrifty Couples" (Cluster 0)

- **Profile**: 905 customers (40.5%). Moderate/low income (\$39.7k), moderate spending (\$222). 100% partner ke sath rehte hain aur ghar mein bacche hain (avg 1.24). Monthly web visits high hain (6.31) aur deals purchases bhi high hain (2.59).
- **Behavior**: Family responsibilities zyada hain isliye disposable income kam hai. Budget stretch karne ke liye discounts dhundhte hain.
- **Marketing Strategy**:
    1. Family bundle discounts aur value packs ("Buy 1 Get 1 Free").
    2. Grocery aur household essentials par seasonal coupon codes.

Persona 1: "Affluent Couples / Established Spenders" (Cluster 1)

- **Profile**: 534 customers (23.9%). Highest income (\$72.8k) aur highest spending (\$1,237). 100% partner ke sath rehte hain, bacche kam hain (avg 0.51). Physical stores (8.66) aur premium catalogs (5.50) se shopping karte hain.
- **Behavior**: Financial stability aur luxury lifestyle. Discounts par depend nahi karte.
- **Marketing Strategy**:
    1. Exclusive VIP Tiered Loyalty Program.
    2. Premium product categories (imported wines, gourmet food) ke personalized catalogs.

Persona 2: "Budget Singles / Window Shoppers" (Cluster 2)

- **Profile**: 444 customers (19.9%). Sabse kam income (\$37.0k) aur sabse kam spending (\$166). Almost 100% akele rehte hain (single parents with avg 1.27 kids). Sabse zyada website visits (6.66/month) lekin actual purchases sabse kam.
- **Behavior**: Window shoppers jo site par aate hain par budget constraints ki wajah se buy nahi kar pate.
- **Marketing Strategy**:
    1. Clearance flash sales aur budget essentials.
    2. Buy Now Pay Later (BNPL) aur zero-interest installment options.
    3. Cart abandonment discount vouchers.

Persona 3: "Affluent Singles / Golden Cluster" (Cluster 3)

- **Profile**: 353 customers (15.8%). High income (\$70.7k), high spending (\$1,190). 100% living alone, minimal kids (0.46). **Highest Campaign Response Rate (32.0%)** jo baki sabhi clusters se double hai!
- **Behavior**: Single high-earning individuals with massive disposable income. Brand ke marketing campaigns ko sabse actively respond karte hain. **Best ROI Cluster**!
- **Marketing Strategy**:
    1. 1-on-1 personalized recommendations aur VIP Concierge service.
    2. High-margin luxury impulse drops aur targeted app notifications.

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3. Project Summary

Is minor project mein humne end-to-end complete ML system design kiya:

1. Missing income median imputation se handle ki.
2. Domain features (`Age` , `Tenure` , `Total_Spending` , `Total_Children` , `Living_With`) engineer kiye.
3. Redundant columns drop kiye aur 4 outliers filter kiye.
4. OneHotEncoder (bina drop first ke) aur StandardScaler apply kiya.
5. 3D PCA se variance visual analysis kiya.
6. KneeLocator aur Silhouette score se $K = 4$ prove kiya.
7. Agglomerative Hierarchical Clustering deploy ki.
8. 4 distinct commercial personas build karke targeted marketing playbooks design kiye.